

Data Cleaning and Preprocessing Report

Medical Appointment No Shows Dataset

Prepared By

Name: Sudha R

Role: Data Analyst Intern

Tool Used: Kaggle Notebook (Python & Pandas)

1. Introduction

Data cleaning and preprocessing are essential steps in the data analysis process. Raw datasets often contain inconsistencies, duplicate records, incorrect data types, and formatting issues that can affect the accuracy of analysis. This project focuses on cleaning the **Medical Appointment No Shows** dataset to improve its quality and prepare it for further exploratory data analysis and visualization.

2. Objective

The primary objectives of this project are to:

- Import the dataset.
 - Examine the dataset structure.
 - Identify missing values.
 - Detect and remove duplicate records.
 - Standardize categorical text values.
 - Convert date columns into appropriate datetime format.
 - Validate numerical values.
 - Save the cleaned dataset for future analysis.
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3. Dataset Description

The **Medical Appointment No Shows** dataset contains records of patients who scheduled medical appointments. It includes demographic information, health conditions, appointment details, and whether the patient attended the appointment.

Dataset Columns

- Patient ID
 - Appointment ID
 - Gender
 - Scheduled Day
 - Appointment Day
 - Age
 - Neighbourhood
 - Scholarship
 - Hypertension
 - Diabetes
 - Alcoholism
 - Handicap
 - SMS Received
 - No Show
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4. Tools and Technologies

- Python
 - Pandas Library
 - Kaggle Notebook
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5. Data Cleaning Process

Step 1: Dataset Import

The dataset was imported into a Pandas DataFrame for preprocessing.

Step 2: Dataset Inspection

The dataset structure was examined using:

- First five records

- Dataset dimensions
- Column names
- Data types

This helped understand the dataset before cleaning.

Step 3: Missing Value Analysis

The dataset was checked for missing values using:

```
df.isnull().sum()
```

Observation

No missing values were found in any column.

Step 4: Duplicate Record Detection

Duplicate records were identified using:

```
df.duplicated().sum()
```

Duplicate rows were removed where applicable to ensure data uniqueness.

Step 5: Standardization

The Gender column values were standardized:

- M → Male
- F → Female

This improves readability and consistency.

Step 6: Date Conversion

The following columns were converted to datetime format:

- ScheduledDay
- AppointmentDay

Datetime conversion enables accurate date-based analysis.

Step 7: Data Validation

The Age column was checked for invalid values.

Records with negative ages were removed if present.

Step 8: Saving the Cleaned Dataset

The cleaned dataset was exported as:

cleaned_medical_appointments.csv

6. Data Quality Improvements

The following improvements were made:

- Imported the dataset successfully.
 - Examined dataset structure.
 - Verified that no missing values existed.
 - Removed duplicate records where necessary.
 - Standardized gender values.
 - Converted date columns to datetime format.
 - Validated age values.
 - Exported a cleaned dataset for further analysis.
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7. Results

After preprocessing, the dataset became cleaner, more consistent, and ready for:

- Exploratory Data Analysis (EDA)
- Data Visualization
- Dashboard Development

- Machine Learning
 - Predictive Analytics
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8. Conclusion

The data cleaning and preprocessing task was completed successfully. The dataset now follows consistent formatting, contains validated records, and is suitable for further analytical tasks. Proper preprocessing improves data quality, enhances reliability, and ensures accurate analytical outcomes.

References

- Kaggle – Medical Appointment No Shows Dataset
- Python Documentation
- Pandas Documentation